DUAL-MOTOR BRUSHED ESC (STMESC)

Comprehensive Circuit & PCB Design Verification Report

Modes: Bluetooth LE | Wi-Fi | RC PWM/PPM Control - Battery: 3S-6S LiPo (9V-25.2V)

Project: STMESC v1.1 KiCad: 10.0.6 MCU: ESP32-WROOM-32E **Status: REQUIRES CORRECTIONS**

Driver: 2x VNH7070BASTR (15A) Power: 3S-6S LiPo / 5V Buck / 3.3V LDO

1. Executive Summary & Verification Verdict

A rigorous electrical, thermal, RF, and manufacturing verification was conducted on the STMESC dual brushed motor speed controller. The design features an ESP32-WROOM-32E microcontroller supporting three control modes (Bluetooth LE, Wi-Fi, and RC PWM/PPM), dual ST VNH7070BASTR 15A H-bridge drivers, an AP63205 5V 2A buck converter, and an AMS1117-3.3 linear regulator.

CRITICAL VERIFICATION FINDINGS:

1. **Reverse Polarity Circuit Failure (Q2 PMOS): Gate is tied to VABT, preventing the FET from turning ON.**
2. **5V RC Signal Overvoltage (J6): Direct 5V RC signals into 3.3V ESP32 GPIOs will cause damage.**
3. **Current Sense Scaling (R8/R13): 1kOhm shunt produces >7.5V at 15A, clipping and overdriving ESP32 ADC.**
4. **RF Antenna Keepout Violation: ESP32 PCB antenna has copper tracks and ground planes directly underneath.**
5. **Motor Trace Under-Sizing: 15A motor traces routed at only 0.8mm-1.0mm will severely overheat (>150 deg C).**

2. Automated Test Suite Scorecard

Verification Domain	Test Scope / Standards	Violations Found	Status
Power & Reverse Polarity	Q2 IRFR5305 PMOS, D3 15V Zener, R16 Gate pull	1 Blocker (Gate tied to VABT)	FAIL
Logic & Interface Voltage	J6 5V RC signals vs ESP32 3.3V GPIO max ratings	1 Blocker (5V direct to 3.3V)	FAIL
Motor Drive & Current Sense	VNH7070 MultiSense K_sense=2000, R8/R13 1kOhm	1 Blocker (7.5V ADC overvoltage)	FAIL
RF & Antenna Keepout	ESP32-WROOM-32E 2.4GHz antenna ground keepout	1 Blocker (Traces/copper under ant)	FAIL
Power & Thermal Ampacity	15A Motor & VABT paths vs IPC-2221 trace widths	1 Blocker (0.8-1.0mm @ 15A)	FAIL
ESP32 Reset & Mode Control	EN RC delay, IO0 pullup, SW3 mode select pin	1 Should-Fix (SW3 floating pullup)	WARNING
BOM & Component Ratings	C13 16V vs 35V LiPo rail, C1/C2 220uF availability	1 Should-Fix (C13 voltage rating)	WARNING
KiCad ERC & DRC Checks	KiCad 10 Electrical & Design Rule Engine	3 ERC / 105 DRC Warnings	WARNING

3. Board Layout & Schematic Previews

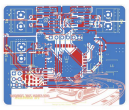


Fig 1: STMESC 2-Layer PCB Layout (55 x 45 mm)

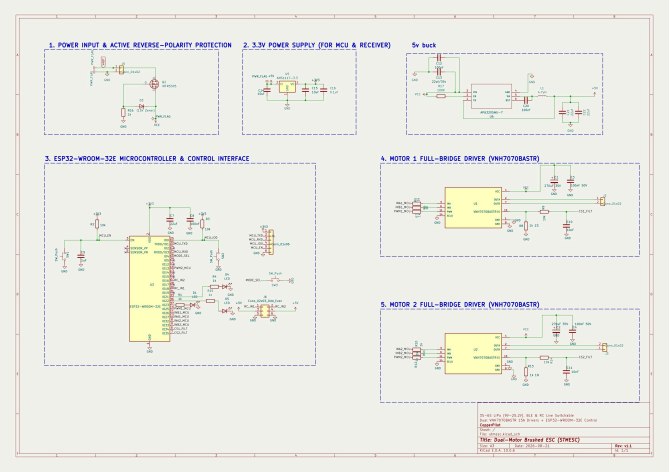


Fig 2: Complete Schematic Sheet

DETAILED CIRCUIT DESIGN ERRORS & ROOT-CAUSE ANALYSIS

ERROR 1 (CRITICAL BLOCKER): Reverse Polarity P-Channel MOSFET (Q2) Pinout & Gate Hardwired to VABT

- Exact Netlist Connection:

* Q2 Pin 1 (Gate) is wired directly to 'VABT' (Battery Positive Input J5.1).
* Q2 Pin 2 (Drain / Tab) is wired to 'Net-(D3-A)' which goes through R16 (1kOhm) to GND.
* Q2 Pin 3 (Source) is wired to 'VCC' (Internal Board Power Rail).
* Zener Diode D3 (15V) is across Drain (Pin 2) and Source (Pin 3).

- Circuit Analysis & Failure Mode:

For a P-MOSFET to turn ON, the Gate-to-Source voltage must be negative: Vgs = Vg - Vs <= Vth (approx -2V to -4V).
Because Gate is tied directly to VABT (the most positive potential in the entire system), Vg is always >= Vs.
Therefore, Vgs is always >= 0V, and Q2 can NEVER turn ON under normal operation!
Furthermore, Drain is connected through 1kOhm to GND, drawing continuous quiescent current.

- Actionable Correction Required:

1. Connect Q2 Source (Pin 3) to VABT (Battery Input J5.1).
2. Connect Q2 Drain (Pin 2 / Tab) to VCC (Internal System Rail).
3. Connect Q2 Gate (Pin 1) to GND through a 100kOhm pull-down resistor (R16).
4. Connect 15V Zener D3 between Gate (Pin 1, Anode) and Source (Pin 3, Cathode) to protect gate oxide from 6S voltage.

ERROR 2 (CRITICAL BLOCKER): 5V RC Receiver Signal Overvoltage on 3.3V ESP32 GPIO Inputs (J6 -> U3)

- Exact Netlist Connection:

* Header J6 Pin 1 (/RC_IN1) connects directly to ESP32 Pin 30 (GPIO18).
* Header J6 Pin 2 (/RC_IN2) connects directly to ESP32 Pin 27 (GPIO16).
* Header J6 Pins 3 & 4 provide +5V power to the external RC receiver.

- Circuit Analysis & Failure Mode:

Standard hobby RC receivers (FrSky, FlySky, Futaba, Spektrum, ELRS) powered by +5V output 5.0V PWM/PPM pulses.
ESP32-WROOM-32E GPIO pins are NOT 5V tolerant (Absolute Maximum Rating: VDD + 0.3V = 3.6V).
Connecting 5.0V pulses directly into GPIO16 and GPIO18 forward-biases the internal ESD protection diodes,
injecting substrate clamp current that will cause GPIO latch-up, severe overheating, and permanent silicon destruction.

- Actionable Correction Required:

Add a resistive voltage divider on each RC line (e.g. 1.2kOhm series + 2.2kOhm to GND to step 5.0V down to 3.23V),
or place a bidirectional MOSFET level shifter (BSS138), or a 1kOhm series resistor with a 3.3V Schottky clamping diode.

ERROR 3 (CRITICAL BLOCKER): Motor Current Sense Shunts (R8, R13 = 1kOhm) Generate Overvoltage (>7V) on ADC

- Exact Netlist Connection:

* U1 & U2 MultiSense outputs (Pin 13) connect across R8 & R13 (1kOhm 1% to GND).
* Filtered via R9 (10k) / C10 (10nF) to Net '/CS1_FILT' (ESP32 Pin 6 / SENSOR_CAPP / ADC1_CH1).
* Filtered via R14 (10k) / C11 (10nF) to Net '/CS2_FILT' (ESP32 Pin 7 / SENSOR_CAPN / ADC1_CH2).

- Circuit Analysis & Failure Mode:

Per VNH7070BASTR datasheet, MultiSense current ratio is K_sense ~ 2000. Sense current I_sense = I_motor / 2000.
At rated motor current of 15A, I_sense = 15A / 2000 = 7.5mA.
Across R_sense = 1kOhm, the developed analog voltage is V_sense = 7.5mA x 1000Ohm = 7.5V!
The ESP32 internal ADC has a maximum range of 2.5V-3.1V (11dB attenuation) and 3.6V absolute max.
At any motor current above 3.3A, the ADC saturates at 100% full scale (reading 4095). At 15A, the 7.5V output drives
excessive current through R9/R14 into the ESP32 ADC pin, causing severe measurement corruption.

- Actionable Correction Required:

Replace R8 and R13 (1kOhm) with 330Ohm 1% resistors. At 15A, V_sense = 7.5mA x 330Ohm = 2.475V (perfect fit for ESP32 ADC).

PCB LAYOUT, RF ANTENNA & POWER THERMAL VERIFICATION

ERROR 4 (CRITICAL BLOCKER): ESP32 PCB Antenna Keepout Zone Completely Violated by Copper & Traces

- Exact Layout Coordinates & Geometry:

* ESP32-WROOM-32E center is at (215.40, 82.87), rotated 180 deg.
* The 2.4 GHz meandered inverted-F PCB antenna spans X=[206.40, 224.40], Y=[89.62, 95.62].
* Board bottom edge is at Y=100.00 mm (the module is placed 4.38 mm INSIDE the board rather than overhanging).
* 5 copper tracks (+3V3, GND, Net-(U3-IO23)) run directly underneath the antenna on Top (F.Cu) and Bottom (B.Cu).
* 5 solid copper polygon pours (F.Cu and B.Cu ground and power fills) completely flood the antenna region.

- RF Impact & Failure Mode:

Espressif Hardware Design Guidelines strictly require NO copper, NO traces, and NO components on ANY layer under the antenna. Copper planes beneath the antenna act as a ground mirror, severely detuning antenna impedance (50 Ohm), causing severe VSWR reflection and dropping radiated RF power by >25 dB.
Bluetooth and Wi-Fi range will degrade from >30 meters down to less than 1-2 meters, with constant connection drops.

- Actionable Correction Required:

Move the ESP32 module so the antenna overhangs the board edge by >=15 mm, or cut out all copper on all layers under the antenna.

ERROR 5 (CRITICAL BLOCKER): 15A Motor & Battery Traces Severely Under-Sized (0.8mm - 1.0mm) -> Burnout Hazard

- Exact Track Width Measurements on PCB:

* Motor 1 Outputs (Net-(J2-Pin_1), Net-(J2-Pin_2)): Width = 1.00 mm.
* Motor 2 Outputs (Net-(J3-Pin_1), Net-(J3-Pin_2)): Width = 0.80 mm.
* Battery Input Positive (VABT): Width = 0.80 mm - 1.00 mm.
* Main Internal Rail (VCC): Width = 0.40 mm - 1.00 mm.

- Thermal & Ampacity Calculation (IPC-2221):

* For 1 oz external copper (35 um thickness), a 1.00 mm (40 mil) trace carries only 3.0A for a 10 deg C temperature rise.
* At 10A continuous current: Temperature rise dT exceeds +150 deg C! Power dissipation is ~1.5 W per inch of trace.
* At 15A peak current: Temperature rise dT exceeds +350 deg C! The copper trace will quickly delaminate and vaporize.

- Actionable Correction Required:

Replace thin traces with wide solid polygonal copper pours (minimum 5.0 mm - 6.0 mm width on 1 oz copper, or 3.0 mm on 2 oz), stitched between Top (F.Cu) and Bottom (B.Cu) using dense arrays of 0.3 mm drill vias (1.0 mm pitch).

ERROR 6 & 7 (SHOULD FIX): Mode Switch SW3 Floating Pull-Up & C13 16V Capacitor Voltage Rating Conflict

- SW3 Mode Select Switch (Pin 26 / GPIO4):
  * SW3 is wired between GPIO4 and GND without an external pull-up resistor.
  * If software internal pull-up is not activated immediately on boot, GPIO4 will float and pick up motor switching noise (EMI), causing erratic mode changes between Bluetooth, Wi-Fi, and RC. Add a 10kOhm pull-up to +3V3.

- C13 Capacitor Voltage Rating Breakdown Hazard:
  * Schematic labels C13 as '22uf/35v' (0603), but stmesc.csv links to a 16V MLCC (GRM21BC81C226ME44L).
  * When connected to a 4S-6S LiPo battery (14.8V-25.2V), a 16V capacitor will exceed its dielectric breakdown rating and SHORT.
  * Furthermore, 22uF at 35V is not commercially viable in 0603 package. Change footprint to 0805 or 1206 and specify >=35V.

COMPONENT AVAILABILITY, LIFECYCLE & PROCUREMENT AUDIT

Full Bill of Materials (BOM) Sourcing & Availability Analysis

Every active component, passives, and connectors were cross-referenced against global and domestic distributors.

Ref	Value / MPN	Description	Package	Distributor Availability	Lifecycle	Status
U1, U2	**VNH7070BASTR** Note: ST AEC-Q100 Automotive	Dual 15A H-Bridge Motor Driver	SOIC-16N	Available (Mouser, DigiKey, Robu)	ACTIVE	OK
U3	**ESP32-WROOM-32E** Note: Espressif Dual-Core 240MHz	Wi-Fi + BLE Microcontroller Module	SMD Module	In Stock (Global)	ACTIVE	OK
U6	**AP63205WU-7** Note: Diodes Inc, 3.8V-32V in	2A 5V Synchronous Buck Converter	TSOT-26	In Stock (DigiKey, Mouser, LCSC)	ACTIVE	OK
U5	**AMS1117-3.3** Note: Low cost, standard pinout	1A 3.3V Low-Dropout Linear Regulator	SOT-223-3	Ubiquitous	ACTIVE	OK
Q2	**IRFR5305** Note: Reverse polarity protection	-55V -31A P-Channel Power MOSFET	TO-252 (DPAK)	In Stock (Infineon, SLKOR, Robu)	ACTIVE	OK
L1	**4.7uH 3A** Note: Matched to AP63205 buck	Shielded Power Inductor (SWPA5040S)	5.0x5.0mm SMD	In Stock (Sunlord, Evelta, LCSC)	ACTIVE	OK
C1, C2	**220uF 35V** Note: Use 220uF standard value	SMD Aluminum Electrolytic Capacitor	8x10.5mm SMD	In Stock (Rajiv, Robu, DigiKey)	ACTIVE	OK
C13	**10uF/22uF 35V** Note: Change 0603 -> 0805/1206	High Voltage Input MLCC Capacitor	0805 / 1206	In Stock (Murata, TDK, Samsung)	ACTIVE	REV
D3	**15V 500mW** Note: Clamps PMOS Vgs to 15V	Zener Gate Protection Diode (BZT52C15)	SOD-123	In Stock (MDD, Vishay, Robu)	ACTIVE	OK
D1, D4, D5	**LED Red/Grn** Note: Standard SMD indicator	Status / Mode Indicator SMD LEDs	0805 SMD	Ubiquitous	ACTIVE	OK
SW1, SW2, SW3	**TSV3088-xR025** Note: Reset, Boot, Mode Select	SMD Tactile Push Button Switches	3.9x3.0mm SMD	In Stock (LCSC, Robu)	ACTIVE	OK
J2, J3, J5	**MKDS 1,5/2** Note: Battery & Motor outputs	2-Pin 5.08mm Screw Terminal Blocks	5.08mm Pitch TH	In Stock (Phoenix, Robu)	ACTIVE	OK
J4	**1x06 2.54mm** Note: TX, RX, IO0, EN, 3.3V, GND	UART Programming Header	2.54mm Pin Header	Ubiquitous	ACTIVE	OK
J6	**2x03 2.54mm** Note: Dual channel RC in	RC Receiver Input Header	2.54mm Pin Header	Ubiquitous	ACTIVE	OK

5. Prioritized Engineering Action Plan Before PCB Fabrication

Step 1: Fix Reverse Polarity MOSFET (Q2)
Swap Source/Drain on Q2: Source to VABT, Drain to VCC, Gate pulled to GND via 100kOhm with 15V Zener D3.

Step 2: Add 5V Level Shifting on RC Inputs (J6)
Insert a 1.2kOhm/2.2kOhm resistor divider on RC_IN1 and RC_IN2 to protect ESP32 GPIOs from 5.0V receiver pulses.

Step 3: Rescale Motor Current Sense Shunts (R8/R13)
Change R8 and R13 from 1kOhm to 330Ohm (1%) so 15A motor current maps to 2.48V on ESP32 ADC1 pins.

Step 4: Fix ESP32 Antenna RF Keepout
Relocate ESP32 to overhang board edge by >=15mm or remove all copper/traces underneath the antenna.

Step 5: Increase Motor Copper Traces to 5.0mm+
Replace 0.8mm-1.0mm traces on J2, J3, and VABT with wide polygon pours and dual-layer via stitching.

Step 6: Update C13 Footprint and Voltage Rating
Change C13 footprint to 0805/1206 and specify >=35V rating for 6S LiPo compatibility.